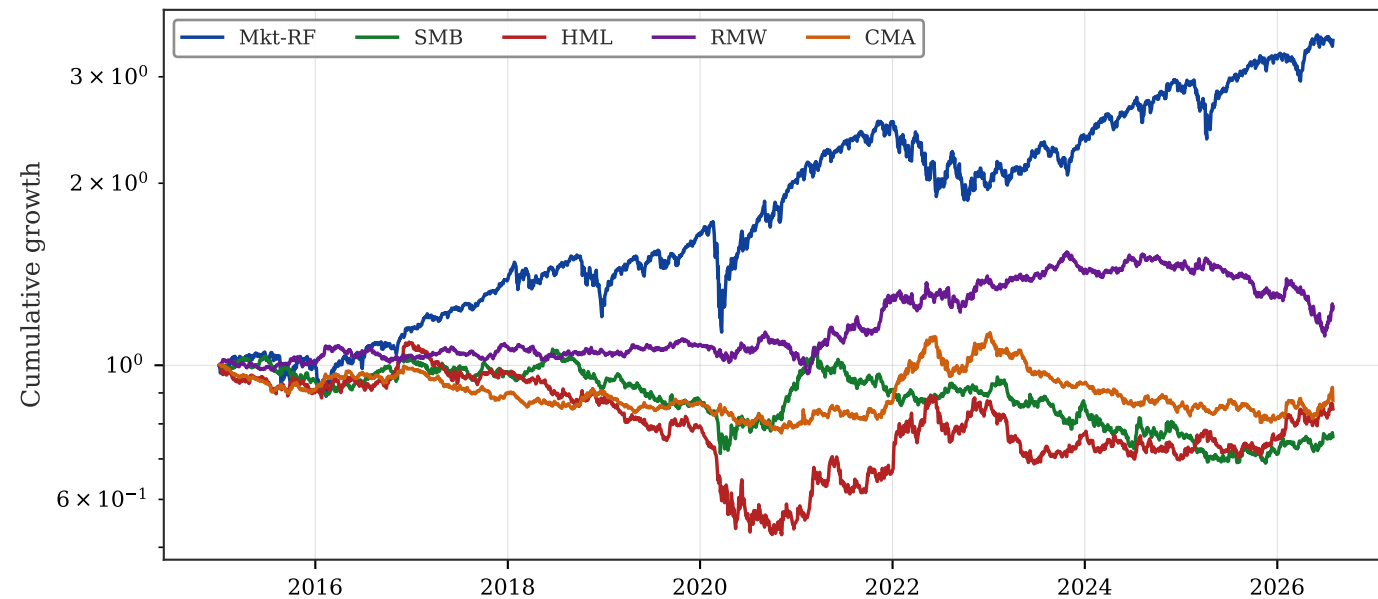
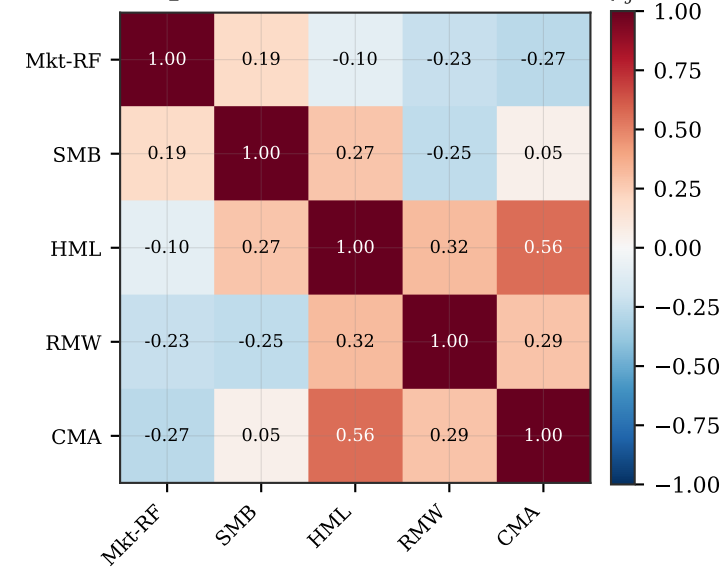


Figure 1 — Data Overview: Fama-French Factors and NOAA Storm-Loss Severity

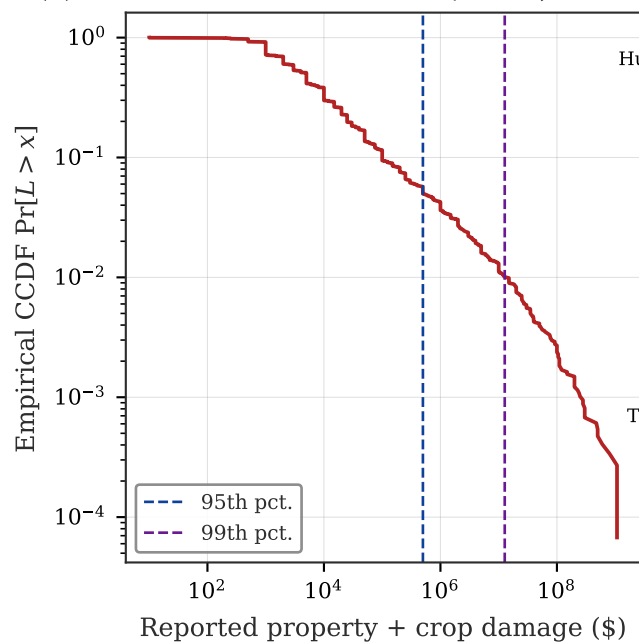
(a) Cumulative Fama-French 5-Factor Growth of \$1 (2015-2026)



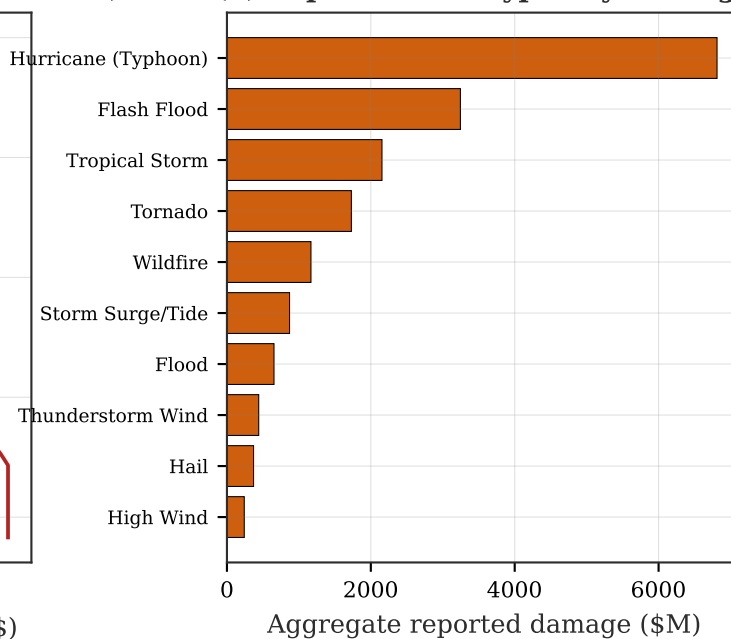
(b) Empirical 5-Factor Correlation ρ_{jk}



(c) NOAA Storm Loss Tail (n=14,784 events)



(d) Top-10 Event Types by Damage



(e) CAT Severity: 64-Bin Log Discretisation

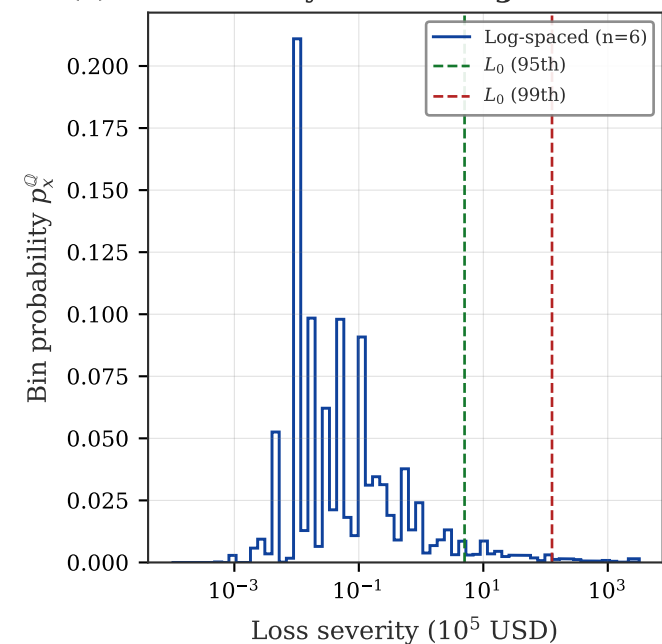
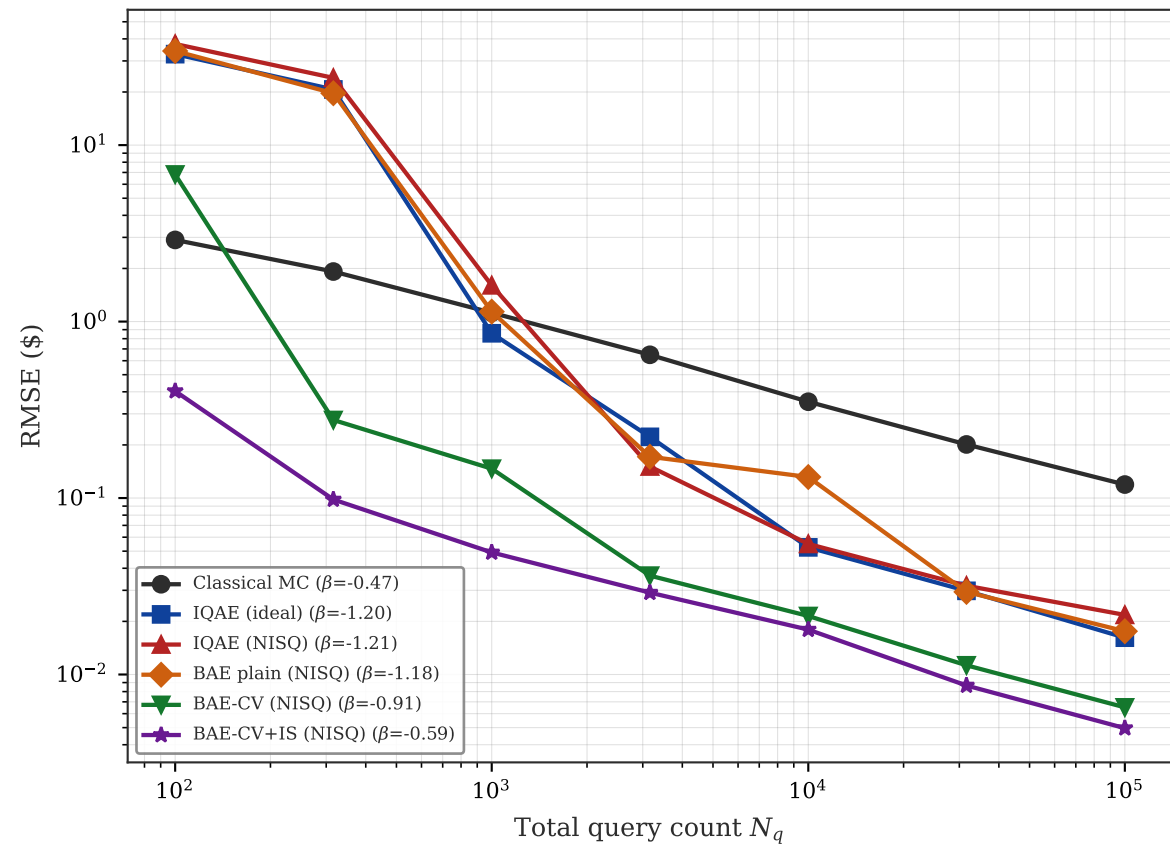


Figure 2 — European Call: RMSE Convergence and Qubit-Count Scaling

(a) European Call: Simulated RMSE Convergence



(b) RMSE / Discretisation Error vs. Qubit Count

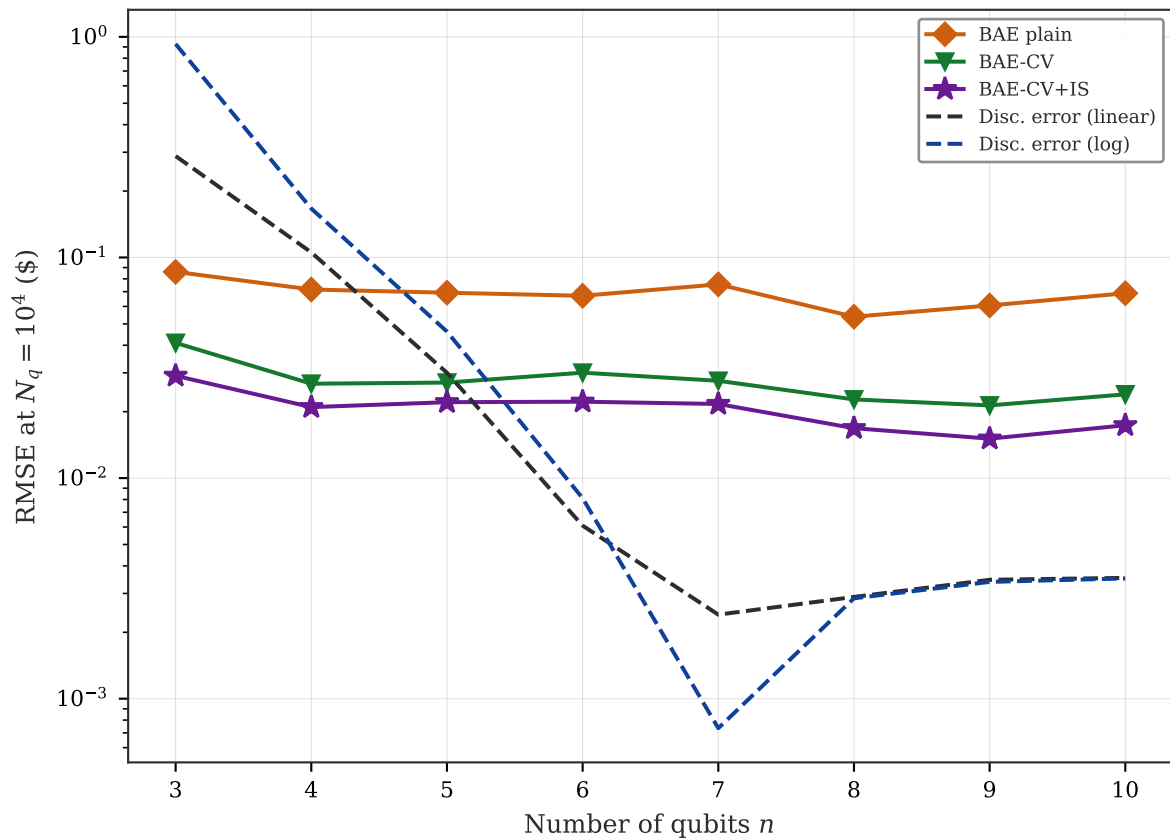
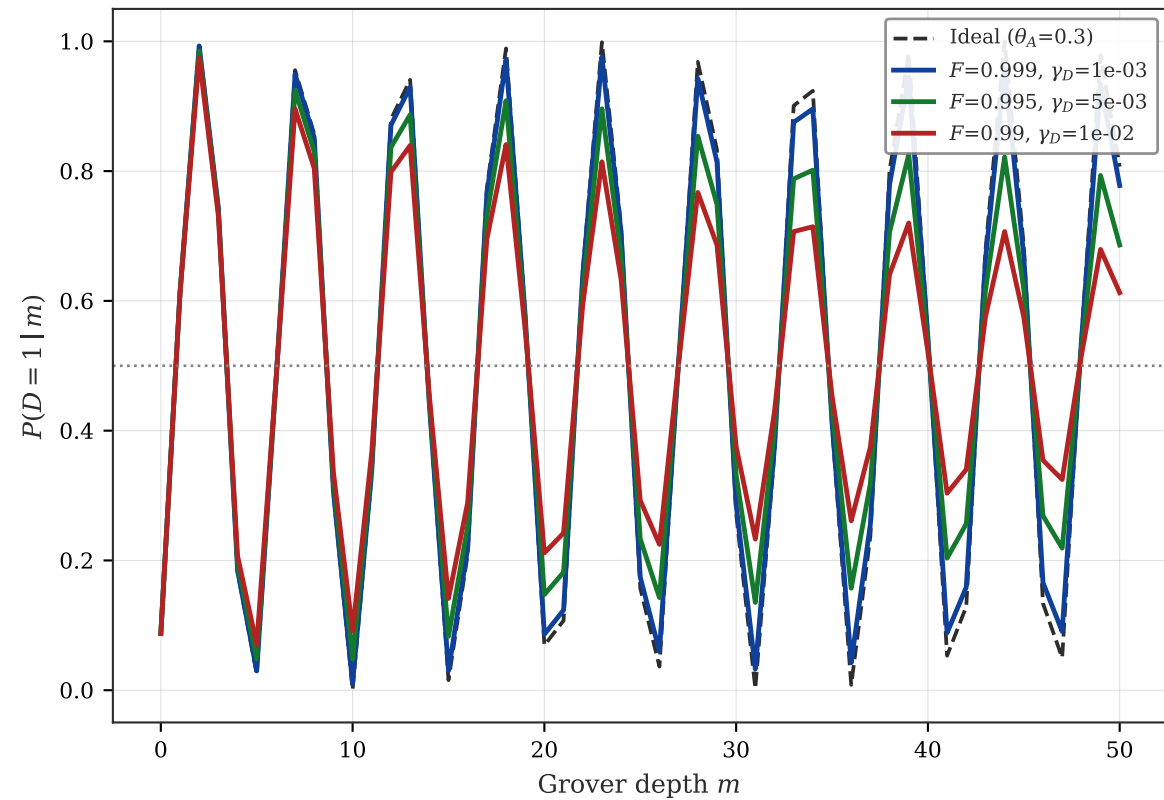


Figure 3 — Bayesian Amplitude Estimation: Noise Model

(a) Noise-Damped Measurement Probability (Eq. noisy_likelihoood)



(b) Noise-Attenuated Fisher Score & Optimal Depth m^*

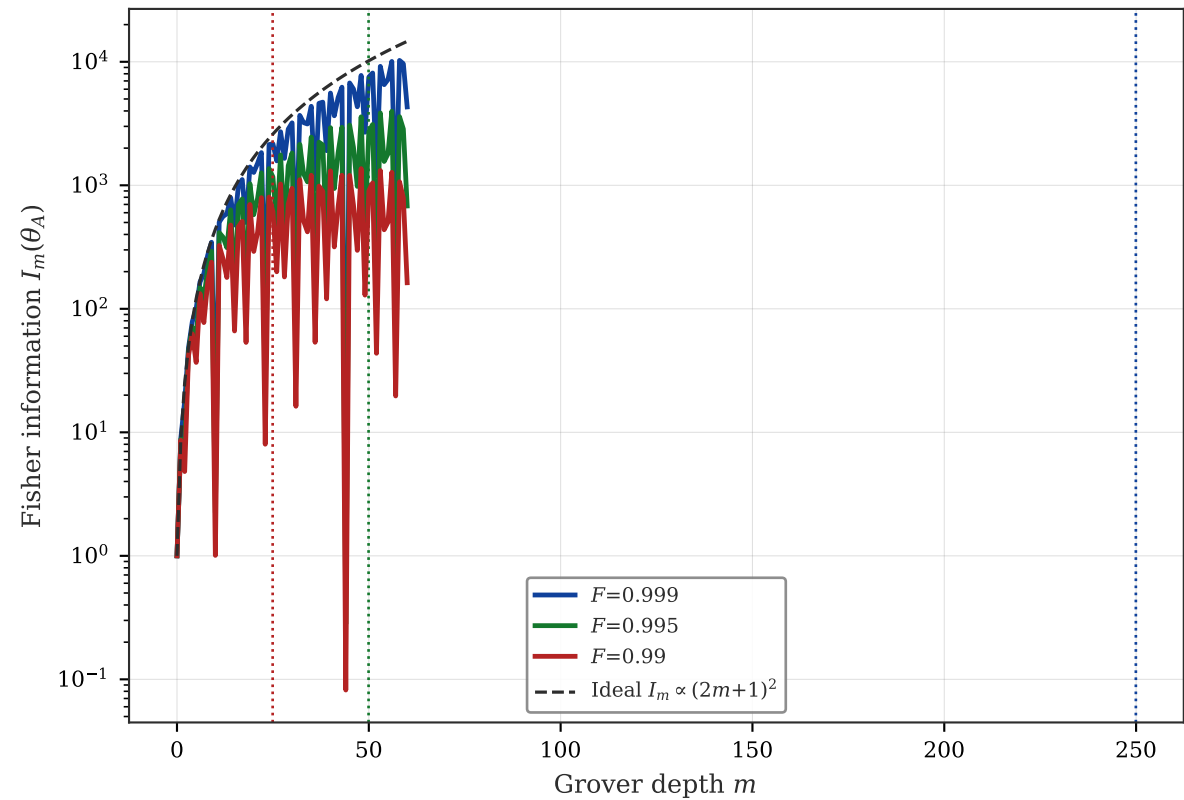


Figure 4 — SMC Posterior Evolution for Joint 3-Parameter Bayesian AE (particle filter, 6000 particles)

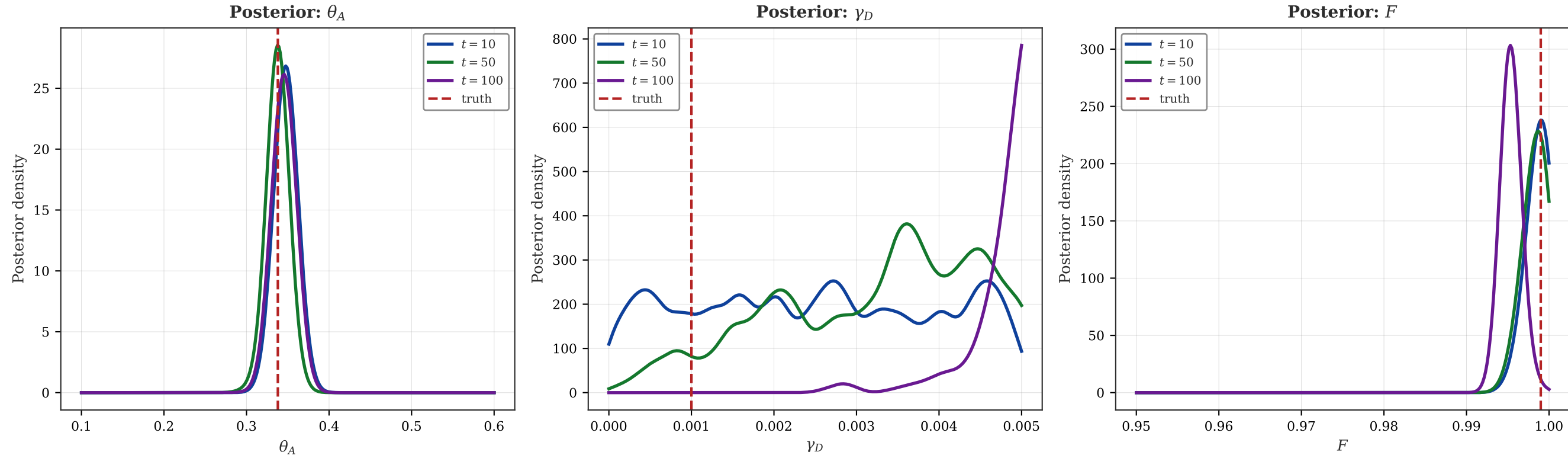
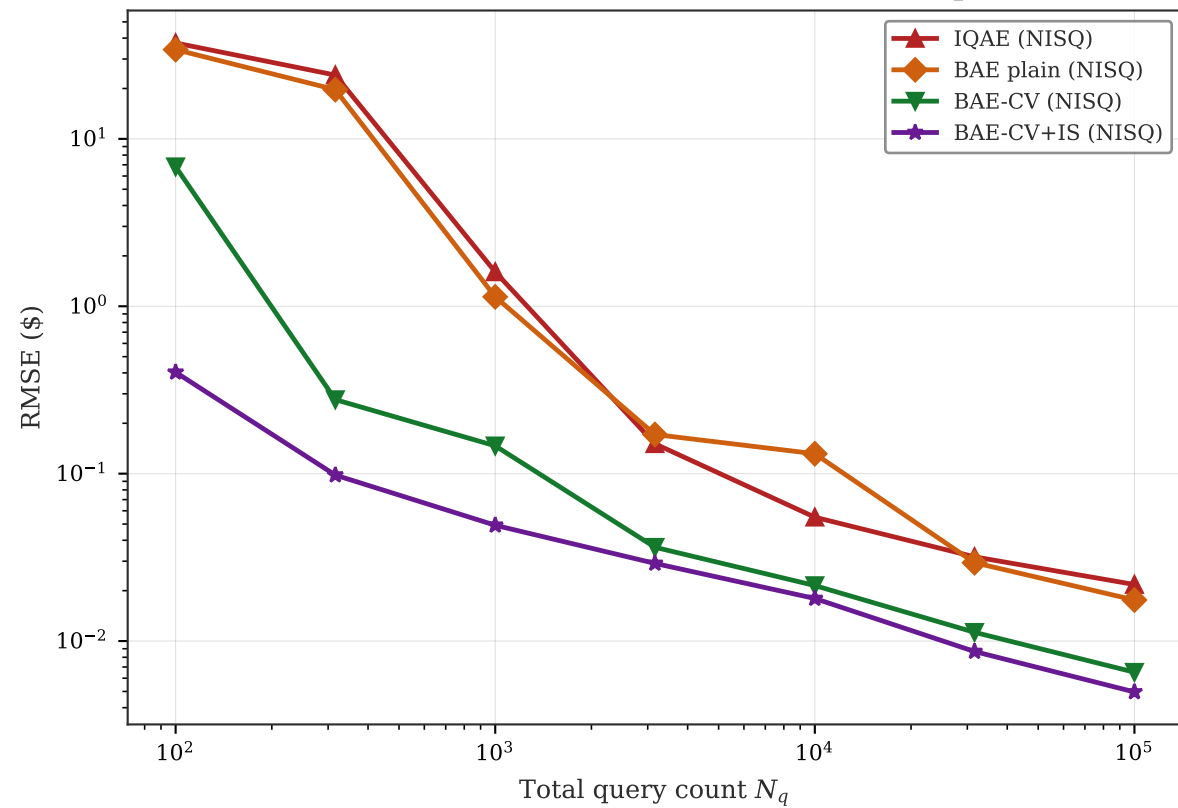


Figure 5 — Quantum Variance Reduction: Empirical Calibration

(a) Effect of Variance Reduction on RMSE (European Call)



(b) CV Query-Reduction Factor vs. Correlation

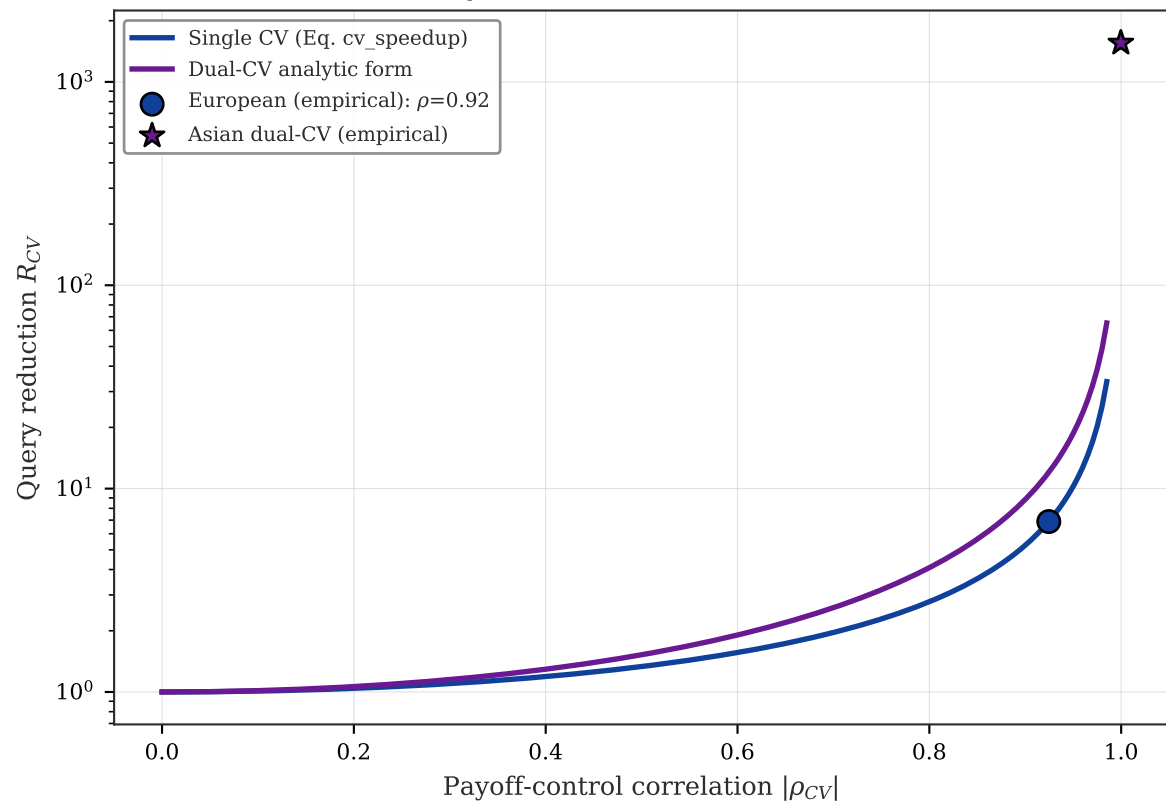
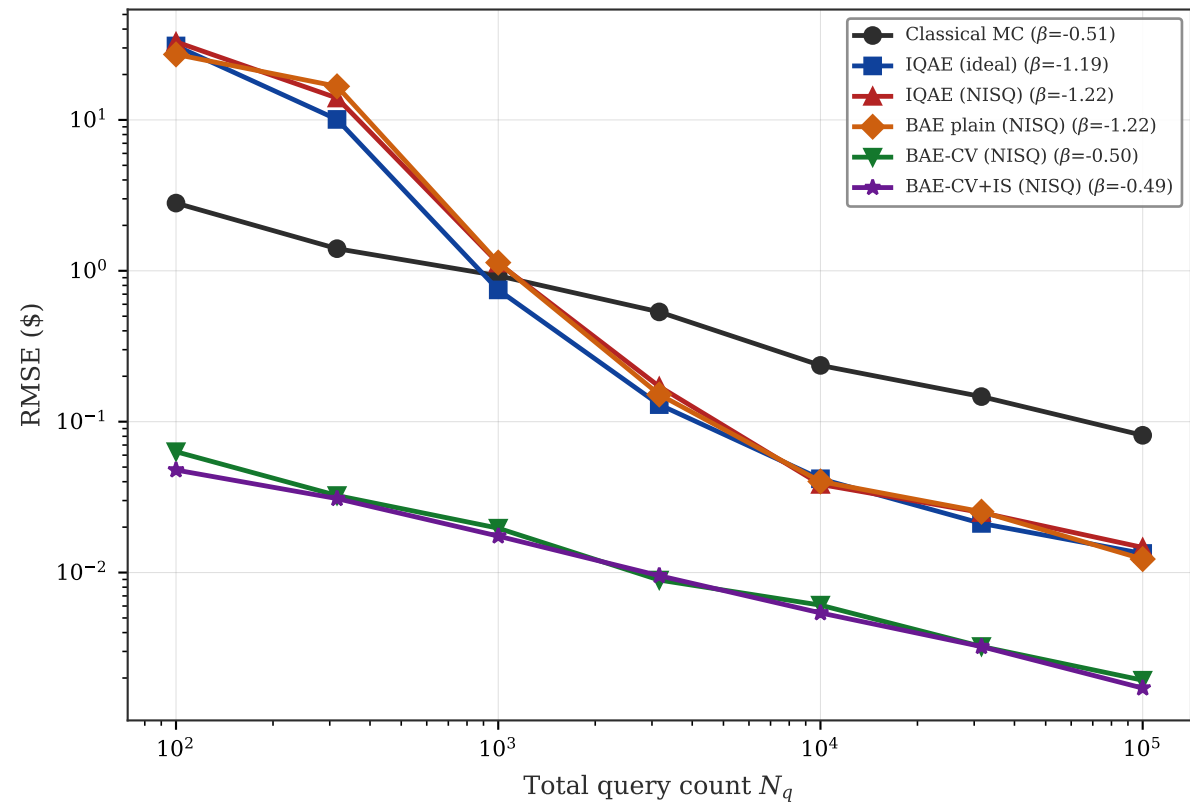


Figure 6 — Asian Call: Dual Control-Variate Analysis

(a) Asian Call RMSE Convergence (12-Step Averaging)



(b) Dual-CV Structure: $\rho_{f, w_1}=0.818$, $\rho_{f, w_2}=1.000$

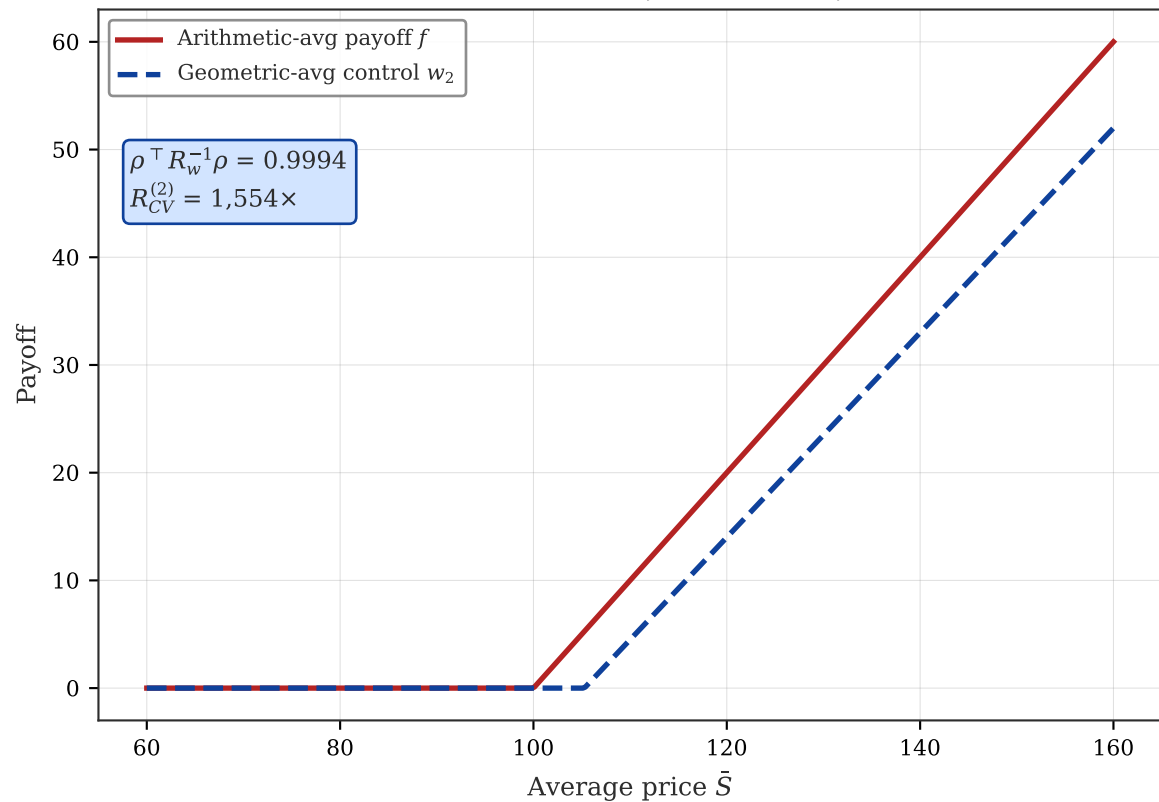
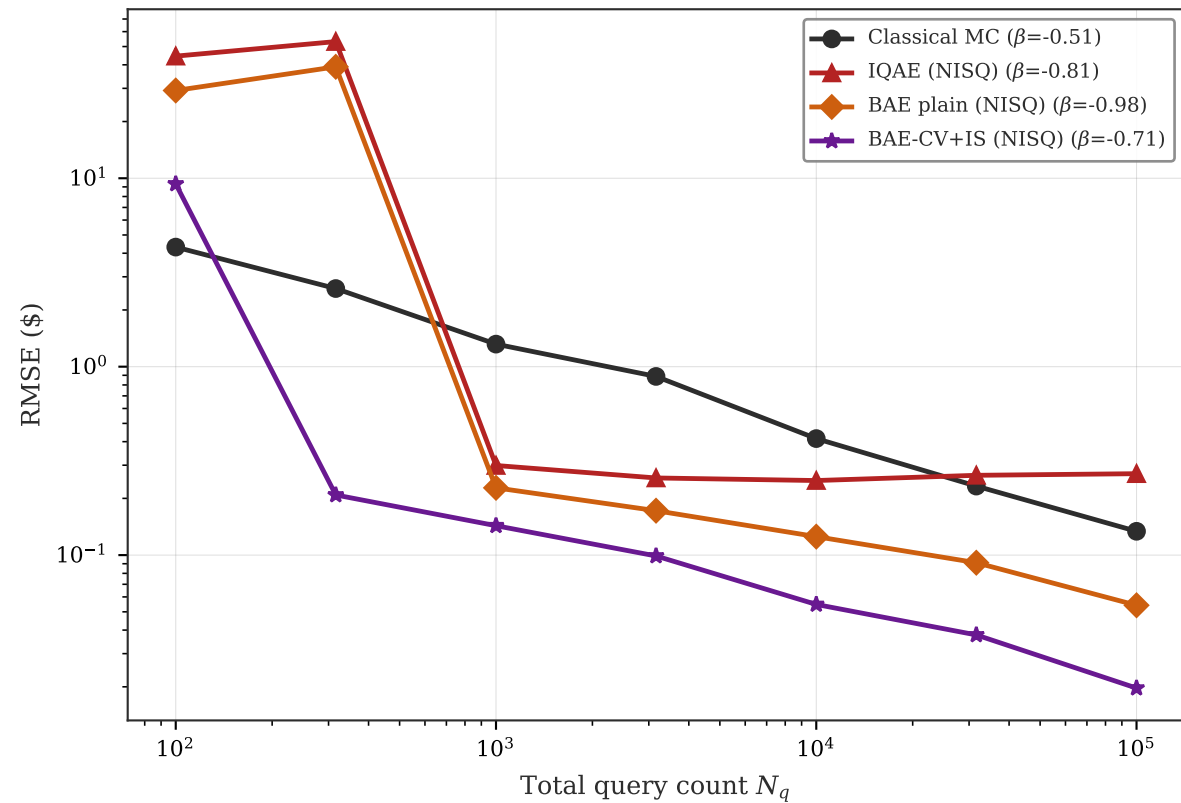


Figure 7 — Multi-Asset Basket Option Benchmarks

(a) $K = 3$ Basket Call RMSE Convergence



(b) Convergence Exponent Erosion vs. Basket Size

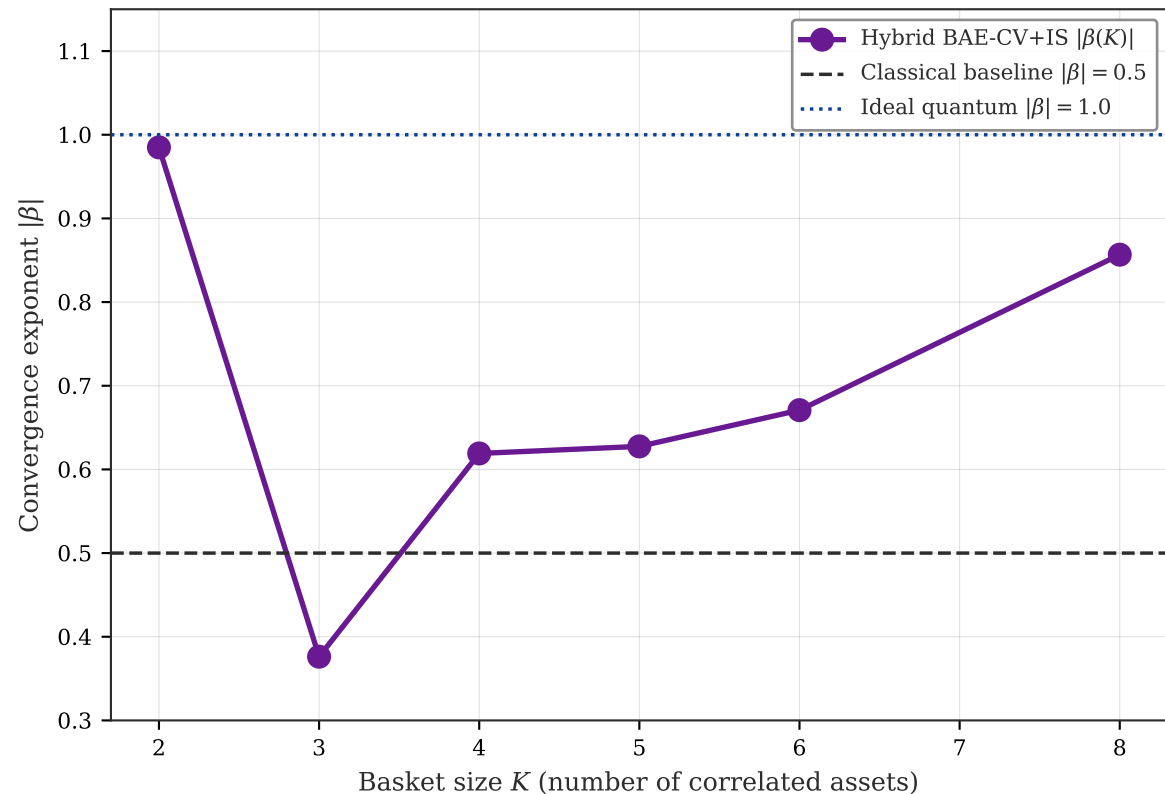
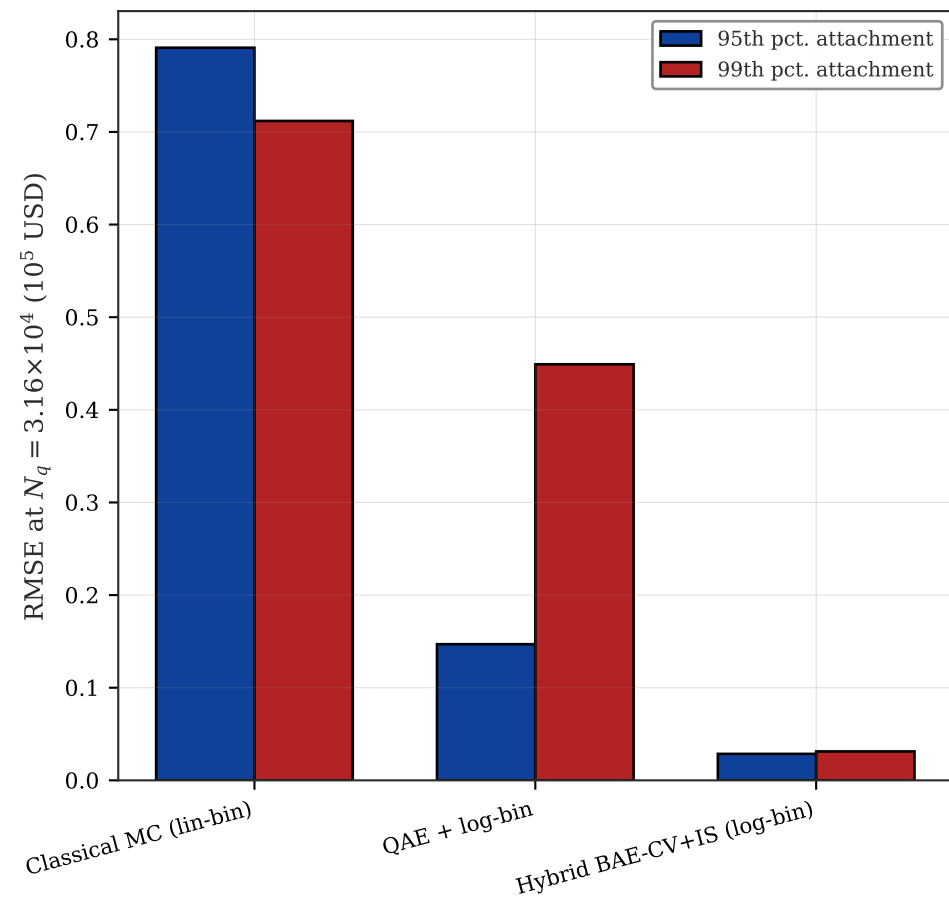


Figure 8 — Catastrophe Tail-Risk Pricing from Real NOAA Storm-Event Data
(a) CAT Tail-Risk RMSE by Method
(n=14,784 real NOAA storm events)



(b) Hybrid BAE-CV+IS Convergence at Both Tail Levels

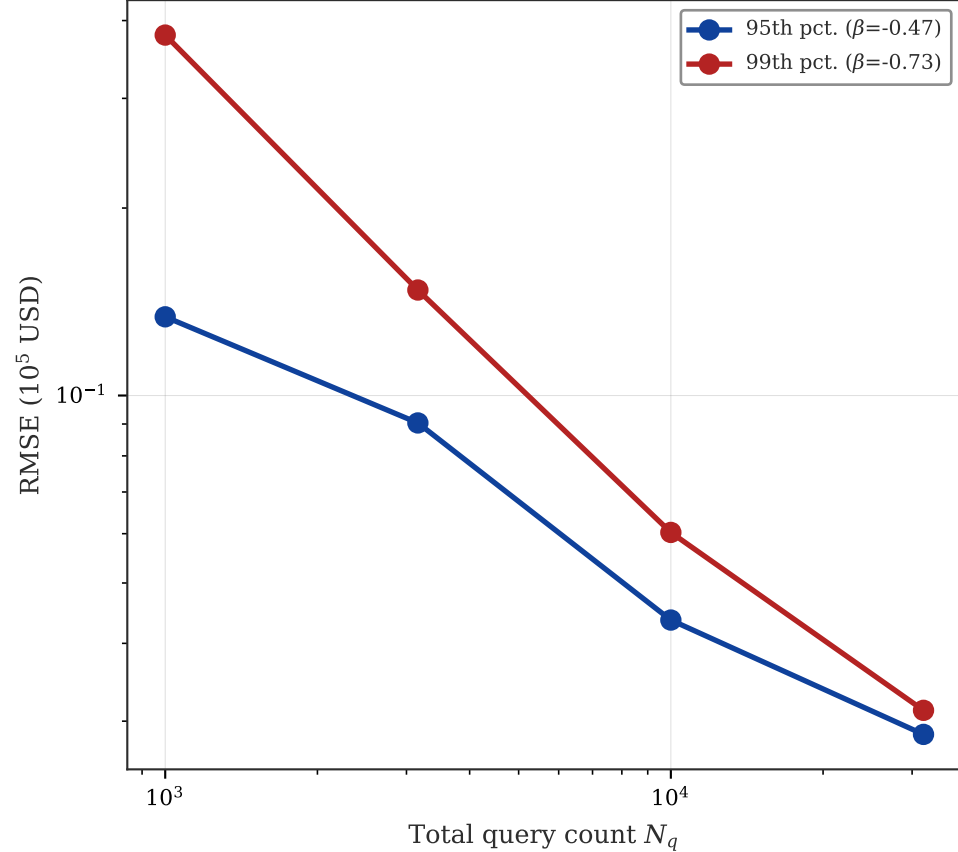
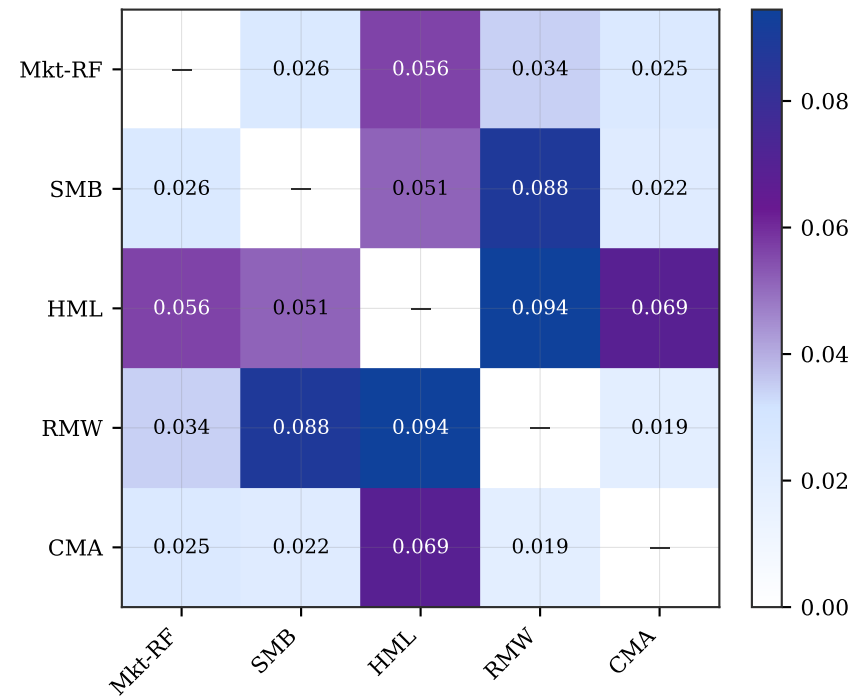
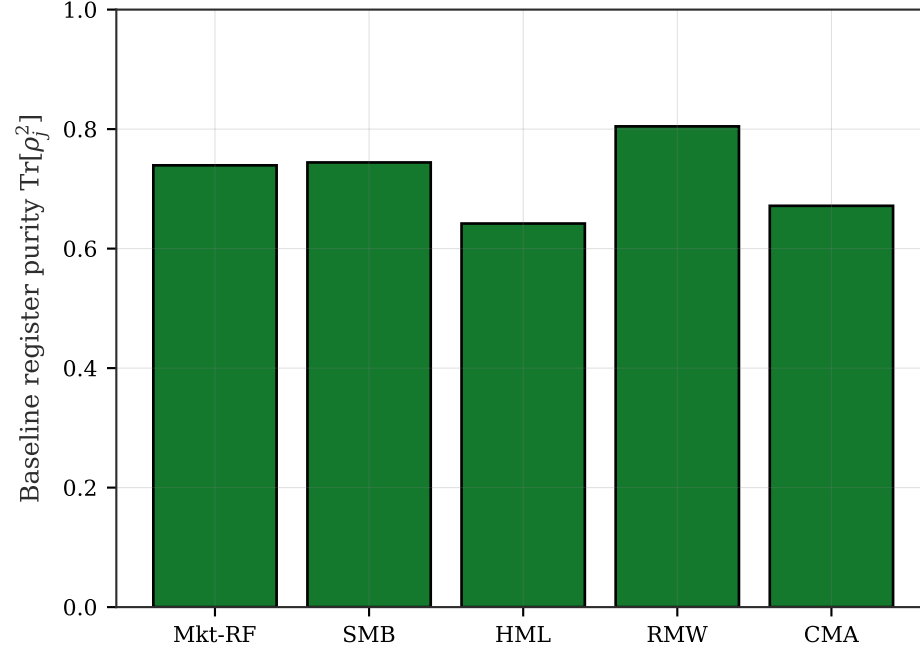


Figure 9 — Quantum Systemic Risk: Entanglement Matrix from Exact Reduced Density Matrices

(a) Systemic Entanglement Matrix E (Eq. qss)
exact reduced-density-matrix computation



(b) Marginal Register Purities



(c) Eigenspectrum of E
Most systemic: HML

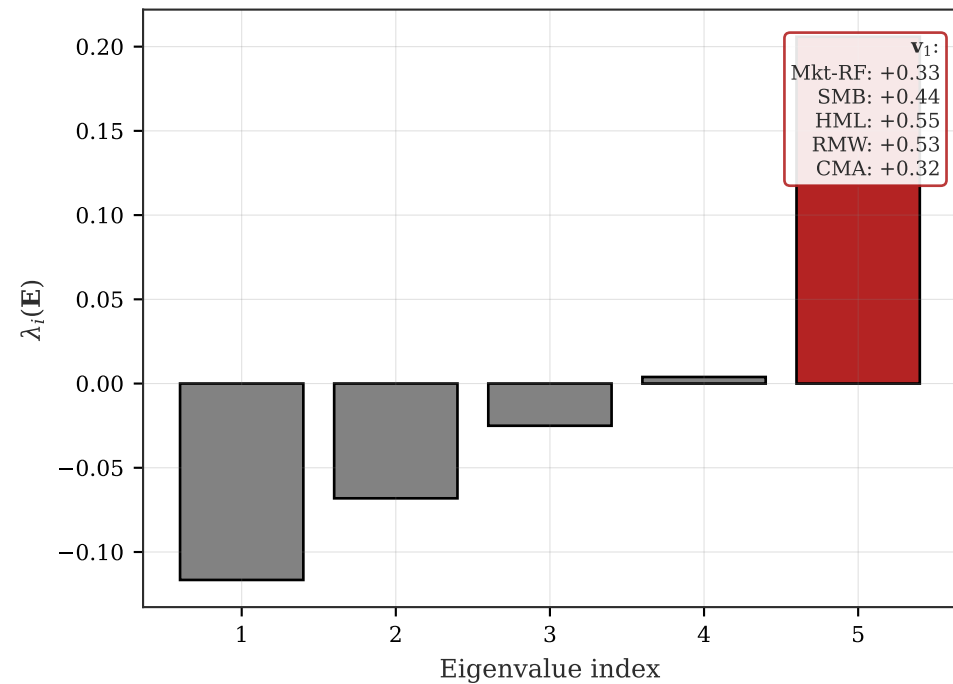
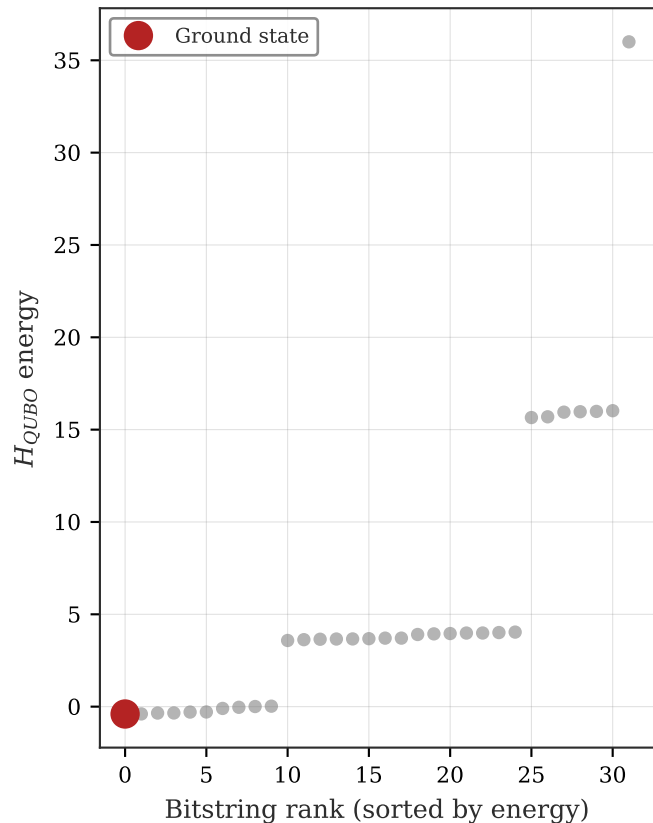
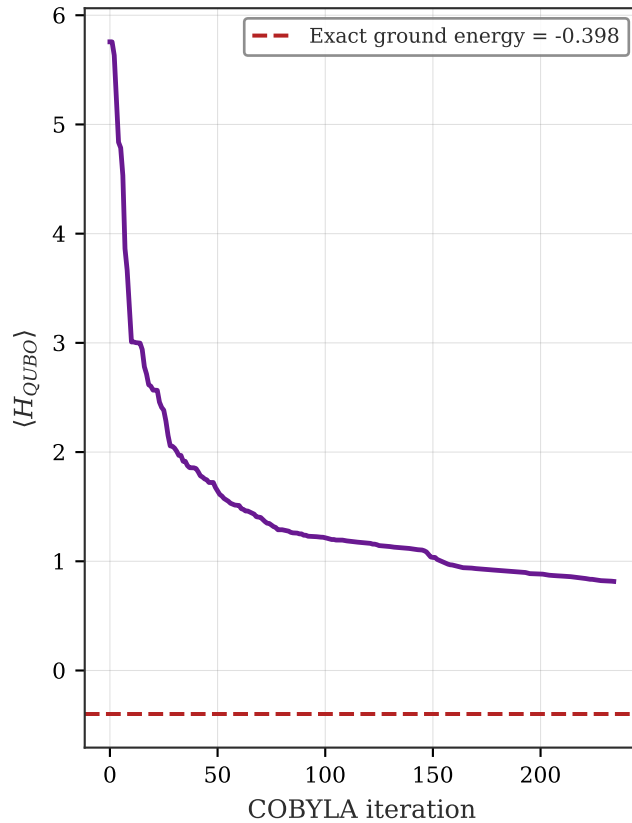


Figure 10 — QUBO Markowitz Portfolio Optimisation: Exact vs. Simulated QAOA

(a) Full QUBO Spectrum ($2^5 = 32$ states)



(b) Simulated QAOA Convergence ($P = 4$ layers)



(c) Top-12 Measured Bitstrings (red = exact optimum)

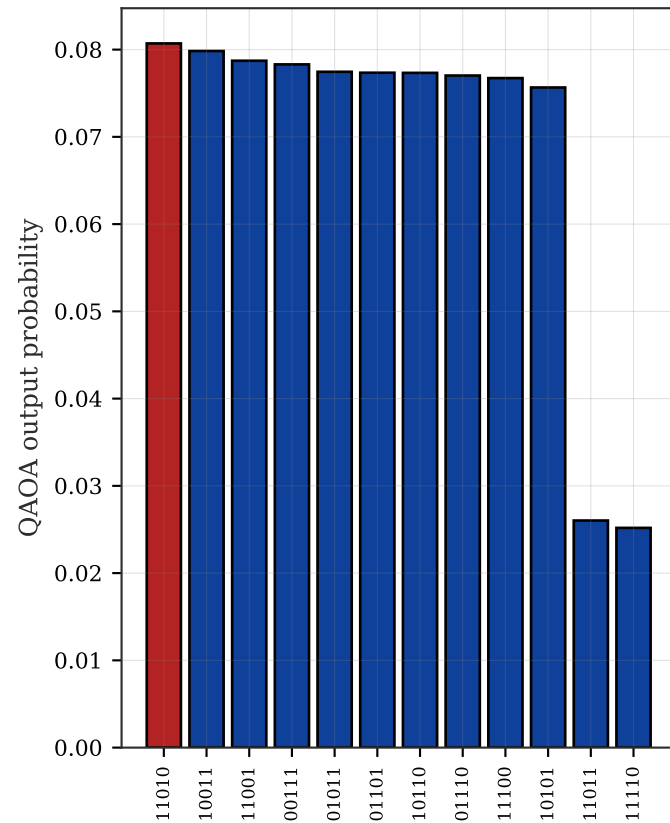
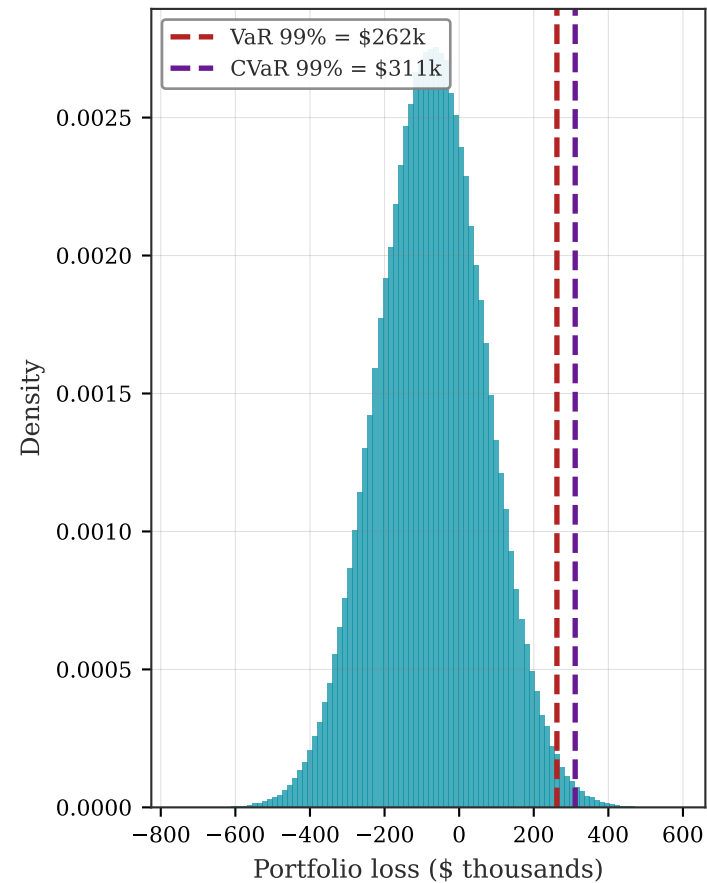
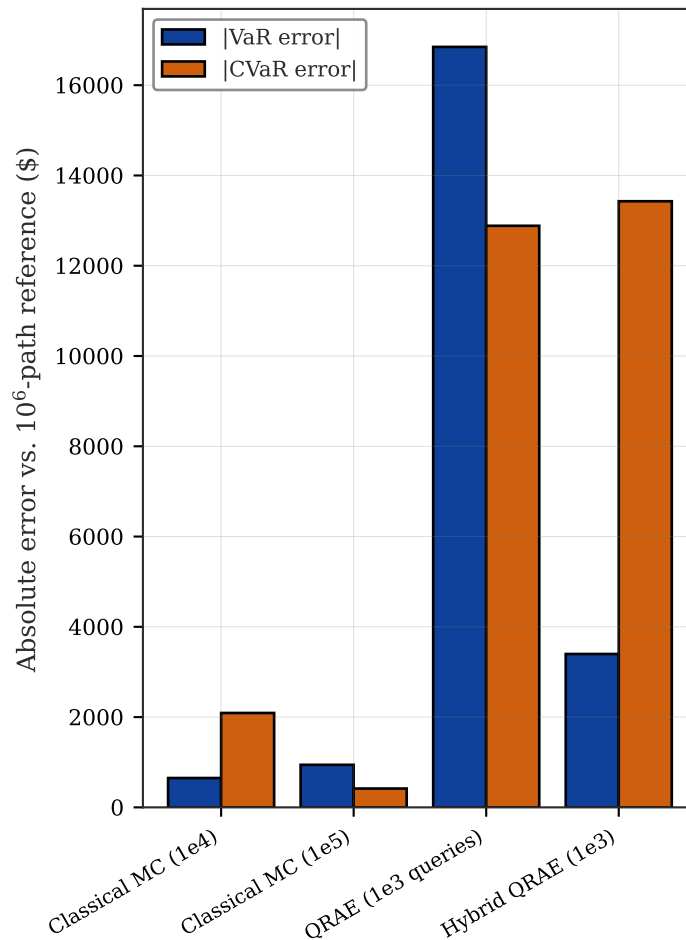


Figure 11 — Quantum Risk Management Framework: VaR/CVaR Results

**(a) 10-Asset Portfolio Loss Distribution
(FF-factor-calibrated Σ)**



(b) VaR/CVaR Estimation Error by Method



(c) VaR/CVaR Risk Profile Across Confidence Levels

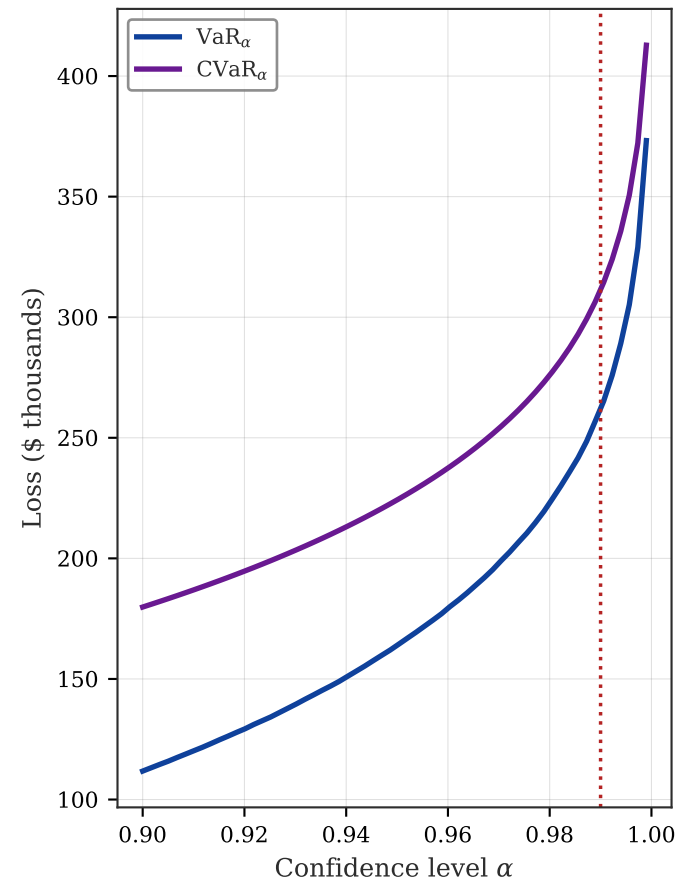
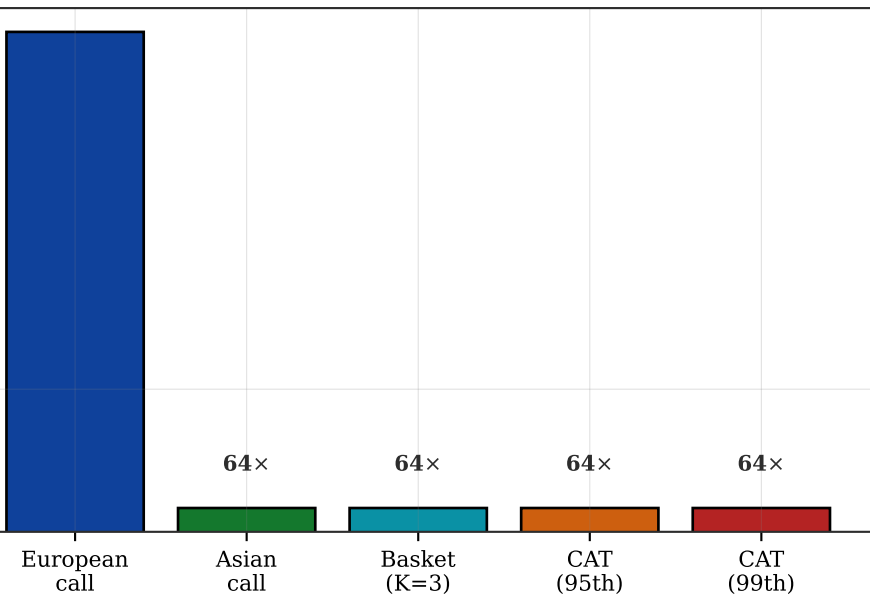
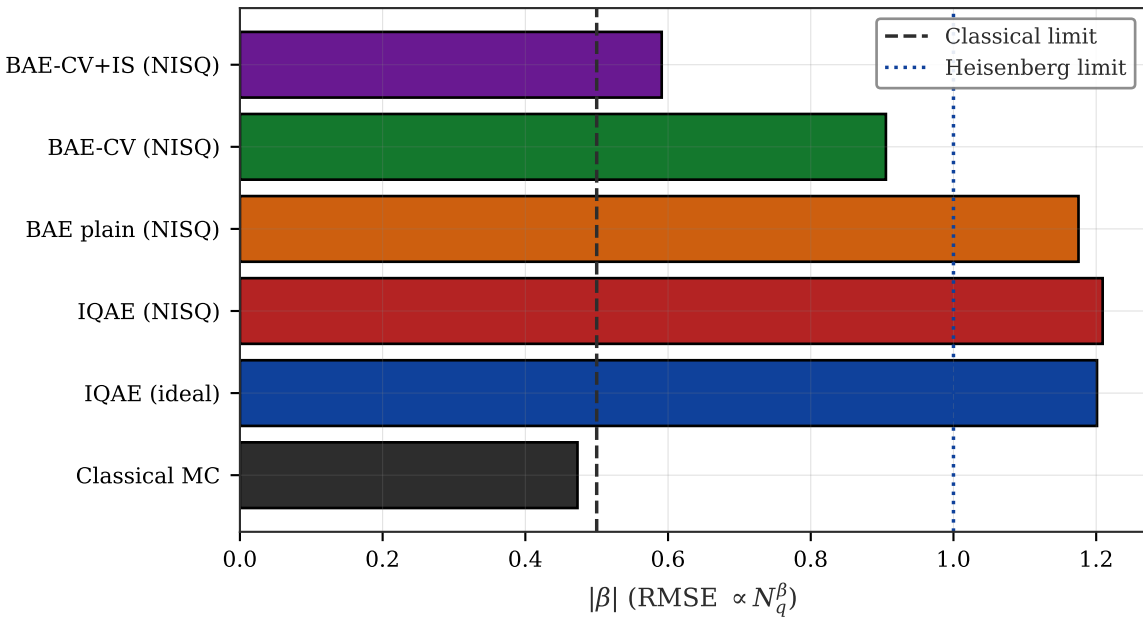


Figure 12 — Summary Dashboard: Hybrid QCMC Results Across All Scenarios

(a) Hybrid BAE-CV+IS Effective Speedup by Scenario



(b) Simulated Convergence Exponents (European Call)



(c) Consolidated QRMF Risk Report

Quantum Risk Management Framework—Key Computed Quantities	
European call (BS closed form)	: \$10.4506
Asian call (12-step MC, 4,000,000 paths)	: \$6.1551
3-asset basket call (Cholesky circuit)	: \$8.3019
CAT excess loss, 95th pct (14,784 events)	: \$8.05 ($\times 10^5$)
CAT excess loss, 99th pct	: \$5.81 ($\times 10^5$)
ρ_{CV} (European, forward control)	: 0.9245 ($R_{CV}=6.89\times$)
Dual-CV Asian $\rho^T R_w^{-1} \rho$	: 0.9994
R_{IS} (European call)	: 1.70x
R_{IS} (CAT, 99th pct.)	: 8.89x
Systemic risk: most systemic factor	: HML
QUBO ground energy (exact)	: -0.3983
QAOA approx. bitstring match to exact	: YES
Hybrid QRAE VaR99% error vs. std. QRAE	: 3398 vs 16848

(d) Query-Complexity Theorem Factor Decomposition (Eq. complexity_theorem, European call)

